



The manufacturer
may use the mark:



Revision 3.1 April 18, 2017
Surveillance Audit Due
May 1, 2020



ANSI Accredited Program
ISO/IEC 17065
PRODUCT CERTIFICATION BODY
#1004

Certificate / Certificat Zertifikat / 合格証

MAR 091051 C001

exida hereby confirms that the:

**Series 99, 91D, 90D, 90, 96D, 88,
83, 77, 22, 88X, 90DX, & 90X
2-Way Ball Valves**

**Mars Valve Co., Ltd.
Taichung, Taiwan – R.O.C.**

Has been assessed per the relevant requirements of:

IEC 61508 : 2010 Parts 1-7

and meets requirements providing a level of integrity to:

Systematic Capability: SC 3 (SIL 3 Capable)

Random Capability: Type A, Route 2_H Device

**PFD_{AVG} and Architecture Constraints
must be verified for each application**

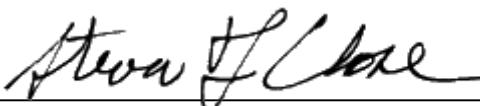
Safety Function:

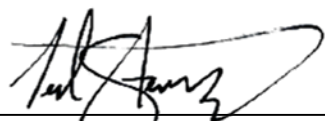
The Ball Valve will move to the designed safe position per the actuator design within the specified safety time.

Application Restrictions:

The unit must be properly designed into a Safety Instrumented Function per the Safety Manual requirements.




Evaluating Assessor


Certifying Assessor

MAR 091051 C001

Systematic Capability: SC 3 (SIL 3 Capable)**Random Capability: Type A, Route 2_H Device****PFD_{AVG} and Architecture Constraints
must be verified for each application****Systematic Capability :**

The product has met manufacturer design process requirements of Safety Integrity Level (SIL) 3. These are intended to achieve sufficient integrity against systematic errors of design by the manufacturer.

A Safety Instrumented Function (SIF) designed with this product must not be used at a SIL level higher than stated.

Random Capability:

The SIL limit imposed by the Architectural Constraints must be met for each element. This Device meets *exida* criteria for Route 2_H.

IEC 61508 Failure Rates, Clean Service in FIT¹

Application	λ_{SD}	λ_{SU}	λ_{DD}	λ_{DU}
Full Stroke	0	0	0	373
Tight Shut-Off	0	0	0	1109
Open on Trip	0	121	0	252
Full Stroke with PVST ²	0	0	126	247
Tight Shut-Off with PVST	0	0	126	983
Open on Trip with PVST	120	1	126	126

¹FIT = 1 failure / 10⁹ hours

²PVST = Partial Valve Stroke Test

SIL Verification:

The Safety Integrity Level (SIL) of an entire Safety Instrumented Function (SIF) must be verified via a calculation of PFD_{AVG} considering redundant architectures, proof test interval, proof test effectiveness, any automatic diagnostics, average repair time and the specific failure rates of all products included in the SIF. Each subsystem must be checked to assure compliance with minimum hardware fault tolerance (HFT) requirements.

The following documents are a mandatory part of certification:

Assessment Report: MAR Q091051 R004 V3R2 or later.

Safety Manual: 25-04-06 Safety Manual 2-Way Ball Valve



80 N Main St
Sellersville, PA 18960

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90, 96D, 88, 83, 77 & 22,
88X, 90DX, 90X 2-Way
Ball Valves